Utkrist Gupta

AI / GenAI Engineer | Python

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Professional Summary

Python AI Engineer with 3 years of experience building Python (FastAPI) backends and LLM-powered applications, including RAG-based systems. Skilled in LangChain, LangGraph, PostgreSQL, and AWS deployment with Docker, Terraform, and CI/CD.

Work Experience

Tata Consultancy Services (TCS) *Bengaluru, Karnataka*

Python AI Engineer *2023 – Present*

Identity Threat Detection & Response (ITDR) *UK-based Banking Client*

- Built a RAG pipeline using pgvector for vector search and BGE-M3 embeddings to ground GPT-4o-based incident analysis in MITRE ATT&CK and MITRE D3FEND knowledge, giving analysts contextual, evidence-backed investigation guidance.
- Designed FastAPI REST services backed by PostgreSQL and SQLAlchemy, exposing relational threat data through well-structured, documented APIs for downstream security modules.
- Built a Python/FastAPI ETL framework to ingest and normalize CSV, XML, and JSON sources into a common schema, loading into PostgreSQL via SQLAlchemy.
- Worked directly with client stakeholders to gather requirements, clarify specifications, and align technical design with business and security priorities.

Cloud Automation & Deployment Platform *US-based Automotive Client*

- Developed Python-based automation for document upload processing in the company portal, reducing manual effort by 73% and improving processing reliability.
- Built and automated Jenkins CI/CD pipelines on AWS EC2 across Development, QA, and Production environments.
- Automated operational tasks, scheduled jobs, and monitoring with Python, Jenkins and AWS Lambda on Linux, cutting manual handling time per task from 30 minutes to under 4 minutes.
- Collaborated with cross-functional teams using Git/GitHub, troubleshooting deployment failures, performing RCA, and improving system reliability.

Personal Projects

LinkedIn Job Agent – Autonomous AI Job Search Agent GitHub

- Built an autonomous job-search agent using LangGraph, LangChain and Gemini/Groq that scrapes listings (JobSpy, Playwright), scores them against a resume, and generates tailored cover letters.
- Implemented deduplication and scheduled execution with automated Telegram alerts, processing 50 jobs per run.

Smart Travel Planner – AI-Powered Travel Planning System Live | GitHub

- Built a full-stack AI trip planner (React/TypeScript, FastAPI) using Groq's LLM API to generate structured, Pydantic-validated itineraries from user inputs like budget and interests.
- Made the system cloud-portable: the same Docker images deploy via Vercel or AWS EKS, provisioned with Terraform and Kubernetes, with Jenkins CI/CD running tests and builds on every push.

Customer Support Triage & Resolution System – Agentic AI Pipeline GitHub

- Built a multi-agent support-ticket pipeline using CrewAI and Groq that classifies tickets, retrieves grounded answers from a ChromaDB knowledge base (RAG), drafts replies, and auto-routes low-confidence cases to human review.
- Designed a code-level confidence gate combining retrieval strength and model self-assessment with input/output guardrails for prompt-injection and PII detection, keeping customer-facing decisions deterministic rather than LLM-judged.

Technical Skills

AI/GenAI: LLM Integration, RAG, Prompt Engineering, LangChain, LangGraph, CrewAI, Gemini API, OpenAI API
Languages: Python, SQL
Backend: FastAPI, REST APIs, SQLAlchemy, Microservices
Databases: PostgreSQL, pgvector, ChromaDB, Redis
Cloud & DevOps: AWS (EC2, S3, VPC, RDS, Lambda, IAM, CloudWatch), Docker, Kubernetes, Terraform, Jenkins, CI/CD, Git/GitHub

Certifications

- AWS Certified Solutions Architect - Associate
- Microsoft Certified Azure Administrator - Associate
- Oracle Cloud Infrastructure Certified Generative AI Professional

Education

Praveer Singh Institute of Technology **2019 – 2023**

Bachelor of Technology in Electronics and Communication Engineering